

# Paper 1 - Set 08 - Solutions

Generated practice paper: answer key and worked solutions.

## Answer key

| Question | Answer | Topic                                      |
|----------|--------|--------------------------------------------|
| 1        | F      | Indices and surds                          |
| 2        | A      | Indices and surds                          |
| 3        | D      | Quadratic functions                        |
| 4        | C      | Quadratic functions                        |
| 5        | B      | Simultaneous equations and inequalities    |
| 6        | D      | Probability                                |
| 7        | C      | Pythagoras, mensuration and 3D             |
| 8        | B      | Functions and transformations              |
| 9        | B      | Simultaneous equations and inequalities    |
| 10       | C      | Pythagoras, mensuration and 3D             |
| 11       | E      | Polynomials, factor and remainder theorems |
| 12       | D      | GCSE algebra                               |
| 13       | C      | Circles and circle theorems                |
| 14       | C      | Trigonometric functions and equations      |
| 15       | A      | Straight lines                             |
| 16       | C      | Exponentials and logarithms                |
| 17       | B      | Exponentials and logarithms                |
| 18       | B      | Straight lines                             |
| 19       | B      | Circles and circle theorems                |
| 20       | B      | Binomial expansion                         |

## Worked solutions

### Question 1 IND-01 · Indices and surds · spec MM1.1

Simplify  $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$ .

**Answer: F**

$(2x^3)^4 = 16x^{12}$ , so the numerator is  $16x^{12} \times 8x^{-5} = 128x^7$ . Dividing by  $4x^2$  gives  $32x^5$ .

#### Full method - how and why

A pure index-laws question. Nothing here needs thought once you treat the numbers and the powers of  $x$  as two separate bookkeeping jobs.

1 Deal with the bracket first:  $(2x^3)^4 = 2^4 x^{12} = 16x^{12}$ .

why: A power outside a bracket multiplies **every** factor inside, so the 2 is raised to the 4th power as well as the  $x^3$ . This is the step people skip.

2 Multiply the numerator:  $16x^{12} \times 8x^{-5} = 128x^7$ .

why: Multiplying powers of the same base adds the indices:  $12 + (-5) = 7$ . The numbers just multiply normally.

3 Divide by  $4x^2$ :  $\frac{128x^7}{4x^2} = 32x^5$ .

why: Dividing subtracts indices:  $7 - 2 = 5$ , and  $128 \div 4 = 32$ .

#### Where it goes wrong

- Forgetting to raise the 2 to the 4th power gives  $2x^{12}$  at step 1 and the answer  $4x^5$ .
- Subtracting instead of adding the  $-5$  gives  $x^9$  and the answer  $32x^9$  - which is offered.

**Takeaway** Handle the number part and the index part separately, and always distribute an outside power over everything in the bracket.

### Question 2 IND-03 · Indices and surds · spec MM1.1

Solve  $4^{x+1} = 8^{2x-3}$ .

**Answer: A**

Write both sides as powers of 2:  $2^{2x+2} = 2^{6x-9}$ . So  $2x + 2 = 6x - 9$ , giving  $4x = 11$  and  $x = \frac{11}{4}$ .

#### Full method - how and why

An exponential equation where both sides can be written as powers of the **same** base. Once the bases match, the indices must be equal.

1 Spot that  $4 = 2^2$  and  $8 = 2^3$ .

why: Both numbers are powers of 2, so the whole equation can be rewritten in base 2 and the exponentials compared directly.

2 Rewrite:  $4^{x+1} = 2^{2(x+1)} = 2^{2x+2}$  and  $8^{2x-3} = 2^{3(2x-3)} = 2^{6x-9}$ .

why:  $(a^m)^n = a^{mn}$ : the outer index multiplies the whole inner index, so brackets are essential.

3 Equate the indices:  $2x + 2 = 6x - 9$ , so  $4x = 11$  and  $x = \frac{11}{4}$ .

why:  $2^u = 2^v$  forces  $u = v$  because  $2^x$  is a one-to-one function (it never takes the same value twice).

#### Where it goes wrong

- Multiplying out as  $3 \times 2x - 3$  instead of  $3(2x - 3)$  loses the bracket and gives a different linear equation.
- Writing  $8 = 2^4$  (a slip) gives  $x = \frac{7}{2}$ , which is offered.

**Takeaway** Whenever an equation has different bases, look for a common base first; comparing indices is then legitimate because exponentials are one-to-one.

### Question 3 QUA-10 · Quadratic functions · spec MM1.3

The quadratic  $x^2 + px + q$  has minimum value  $-4$  and one of its roots is  $0$ . Which of the following could be the quadratic?

**Answer: D**

Minimum  $-4$  means  $x^2 + px + q = (x - h)^2 - 4$ , with roots  $h \pm 2$ . One root is  $0$ , so  $h = \pm 2$ : the quadratic is  $x^2 - 4x$  or  $x^2 + 4x$ . Only  $x^2 - 4x$  is offered.

#### Full method - how and why

Two pieces of information about a quadratic - its minimum and one root - pin the whole thing down. Working in completed-square form makes the roots visible.

- 1 Minimum  $-4$  and leading coefficient  $1$  means  $x^2 + px + q = (x - h)^2 - 4$  for some  $h$ .  
why: A monic quadratic with minimum  $m$  must be  $(x - h)^2 + m$ ; only the position  $h$  is unknown.
- 2 Its roots satisfy  $(x - h)^2 = 4$ , i.e.  $x = h \pm 2$ .  
why: Solve the completed-square form directly rather than expanding.
- 3 One root is  $0$ , so  $h + 2 = 0$  or  $h - 2 = 0$ :  $h = -2$  or  $h = 2$ .  
why: Both cases are legitimate; the options decide which one is wanted.
- 4 That gives  $x^2 + 4x$  or  $x^2 - 4x$ ; the list offers  $x^2 - 4x$ .  
why: Check it:  $x^2 - 4x = (x - 2)^2 - 4$ , minimum  $-4$ , roots  $0$  and  $4$ .

#### Where it goes wrong

- Picking  $x^2 - 2x - 3$ : its minimum is  $-4$  but its roots are  $-1$  and  $3$ , so  $0$  is not a root.
- Checking only the minimum (several options have minimum  $-4$ ) or only the roots.

**Takeaway** Test every stated condition - an option that satisfies one of them is designed to catch you.

### Question 4 QUA-16 · Quadratic functions · spec MM1.3

Given that  $x^2 + y^2 = 10$  and  $x + y = 4$ , find the value of  $xy$ .

**Answer: C**

$(x + y)^2 = x^2 + y^2 + 2xy$ , so  $16 = 10 + 2xy$  and  $xy = 3$ .

#### Full method - how and why

A symmetric pair of equations. Expanding  $(x + y)^2$  links the two given quantities to  $xy$  directly.

- 1 Write the identity  $(x + y)^2 = x^2 + y^2 + 2xy$ .  
why: This is the only identity that connects a sum, a sum of squares and a product.
- 2 Substitute the given values:  $4^2 = 10 + 2xy$ .  
why:  $x + y = 4$  and  $x^2 + y^2 = 10$  are both given, so only  $xy$  is unknown.
- 3 Solve:  $16 - 10 = 2xy \Rightarrow xy = 3$ .  
why: Halve at the end - forgetting to divide by  $2$  gives  $6$ , which is offered.

#### Where it goes wrong

- Answering  $6$  by omitting the factor  $2$  in  $2xy$ .
- Trying to solve for  $x$  and  $y$  individually (they are  $1$  and  $3$ ) - slower and unnecessary.

**Takeaway**  $(x + y)^2 = x^2 + y^2 + 2xy$  converts between the three symmetric quantities; recognise which two you are given.

**Question 5** SIM-17 · Simultaneous equations and inequalities · spec MM1.4

Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?

**Answer: B**

$(x - y)^2 = 2(x^2 + y^2) - (x + y)^2 = 116 - 100 = 16$ , so the difference is 4 (the numbers are 3 and 7).

**Full method - how and why**

Symmetric information again: a sum and a sum of squares. The identity for  $(x - y)^2$  gets the difference without finding the numbers.

**1** Use  $(x - y)^2 = x^2 + y^2 - 2xy$  and  $(x + y)^2 = x^2 + y^2 + 2xy$ .

*why:* Adding these gives  $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$  - the relation that links all three given quantities.

**2** Rearrange:  $(x - y)^2 = 2(x^2 + y^2) - (x + y)^2$ .

*why:* This avoids having to find  $xy$  as an intermediate step.

**3** Substitute:  $2(58) - 10^2 = 116 - 100 = 16$ .

*why:* Careful: it is  $2(x^2 + y^2)$ , not  $x^2 + y^2$ .

**4** Take the positive square root: the difference is 4 (the numbers are 3 and 7).

*why:* The question asks for the **positive** difference, so take +4.

**Where it goes wrong**

- Forgetting the factor 2 and getting  $(x - y)^2 = -42$ , which is impossible.
- Giving 16 (the square of the difference) as the answer.

**Takeaway**  $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$ : any two of sum, difference and sum-of-squares give the third.

**Question 6** PRO-09 · Probability · spec M7.4

The probabilities that a train is early, on time or late are 0.1,  $x$  and  $2x$  respectively. Find the probability that the train is not late.

**Answer: D**

$0.1 + 3x = 1 \Rightarrow x = 0.3$ ; not late  $= 0.1 + 0.3 = 0.4$ .

**Full method - how and why**

The probabilities of an exhaustive set of outcomes sum to 1. Use that to find  $x$  first.

**1** Early, on time, late are the only possibilities, so  $0.1 + x + 2x = 1$ .

*why:* Exhaustive and mutually exclusive outcomes have total probability 1.

**2** Solve:  $3x = 0.9$ , so  $x = 0.3$  and late  $= 0.6$ .

*why:* Subtract 0.1, then divide by 3.

**3** Not late  $=$  early + on time  $= 0.1 + 0.3 = 0.4$ .

*why:* Or use the complement:  $1 - 0.6 = 0.4$  - quicker and a useful check.

**Where it goes wrong**

- Stopping at  $x = 0.3$  and answering that, or answering 0.6, the probability of being late.
- Forgetting 'early' counts as 'not late'.

**Takeaway** Probabilities of all outcomes sum to 1; 'not late' is the complement of 'late'.

**Question 7** MEN-16 · Pythagoras, mensuration and 3D · spec M5.18

From the top of a cliff 40 m high, the angle of depression of a boat is  $30^\circ$ . How far is the boat from the foot of the cliff?

**Answer: C**

$$\tan 30^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}.$$

**Full method - how and why**

An angle of depression from the top equals the angle of elevation from the boat. Then it is right-angled trigonometry.

- 1 Draw the triangle: the cliff is vertical (40 m), the sea horizontal, and the angle at the boat is  $30^\circ$ .

why: The angle of depression from the cliff top equals the angle of elevation from the boat (alternate angles with the horizontal).

- 2 Relative to the  $30^\circ$  angle, the cliff is opposite and the distance  $d$  is adjacent:  $\tan 30^\circ = \frac{40}{d}$ .

why:  $\tan = \frac{\text{opposite}}{\text{adjacent}}$ .

- 3 Solve:  $d = \frac{40}{\tan 30^\circ} = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}$  m.

why: Dividing by  $\frac{1}{\sqrt{3}}$  multiplies by  $\sqrt{3}$ .

**Where it goes wrong**

- Multiplying by  $\tan 30^\circ$  instead of dividing, giving  $\frac{40}{\sqrt{3}}$  - an offered option.
- Placing the  $30^\circ$  angle at the top of the cliff inside the triangle, which would swap opposite and adjacent.

**Takeaway** Depression from the top = elevation from the bottom; a shallow angle means a large horizontal distance.

**Question 8** FUN-05 · Functions and transformations · spec MM8.2

The graph of  $y = f(x)$  passes through (4, 6). Which point must lie on the graph of  $y = 3f(2x) - 1$ ?

**Answer: B**

$f(2x)$  halves  $x$ -coordinates: (2, 6). Then  $3f - 1$ :  $y = 18 - 1 = 17$ . Point (2, 17).

**Full method - how and why**

Track a single known point through the transformations, remembering that inside changes act inversely on  $x$ .

- 1 Start with the known point (4, 6) on  $y = f(x)$ .

why: One point is all you are given, so the answer must be its image.

- 2  $f(2x)$  squashes horizontally by factor  $\frac{1}{2}$ : the point with  $x$ -coordinate 4 moves to  $x = 2$ , since  $f(2 \times 2) = f(4) = 6$ .

why: To get the same output from  $f(2x)$  you need half the input - this is the inverse action of an inside change.

- 3 Multiply by 3:  $y = 3 \times 6 = 18$ . Then subtract 1:  $y = 17$ .

why: Outside changes act directly on the output, in the order written.

- 4 The point is (2, 17).

why: Check:  $3f(2 \times 2) - 1 = 3(6) - 1 = 17$  ✓.

**Where it goes wrong**

- Doubling the  $x$ -coordinate to 8 instead of halving it.
- Forgetting the  $-1$  and answering (2, 18).

**Takeaway** Substitute the point back into the new equation to check:  $f(2x)$  at  $x = 2$  really is  $f(4)$ .

**Question 9** SIM-14 · Simultaneous equations and inequalities · spec MM1.4

Solve  $2x + 3y = 12$  and  $x^2 - y^2 = 0$ . How many solutions  $(x, y)$  are there?

**Answer: B**

$x^2 = y^2$  means  $y = x$  or  $y = -x$ .  $y = x$ :  $5x = 12$ , one solution.  $y = -x$ :  $-x = 12$ , one solution. Two in total.

**Full method - how and why**

$x^2 - y^2 = 0$  factorises into two lines. The system is a line meeting a pair of lines, so count the intersections of each case.

- Factorise:  $x^2 - y^2 = (x - y)(x + y) = 0$ , so  $y = x$  or  $y = -x$ .  
why: A product is zero when either factor is zero - this splits the problem into two straight-line cases.
- Case  $y = x$ :  $2x + 3x = 12 \Rightarrow 5x = 12$ , one solution  $(\frac{12}{5}, \frac{12}{5})$ .  
why: Substitute into the linear equation; a linear equation in one unknown has exactly one root.
- Case  $y = -x$ :  $2x - 3x = 12 \Rightarrow -x = 12$ , one solution  $(-12, 12)$ .  
why: Same method; the coefficient of  $x$  is non-zero so again exactly one solution.
- Total: two solutions.  
why: The two cases give different points, so no double counting.

**Where it goes wrong**

- Taking the square root as  $y = x$  only, halving the answer.
- Concluding 'a line meets a quadratic in at most 2 points, so 2' without checking that neither case is parallel or degenerate.

**Takeaway**  $A^2 = B^2$  means  $A = B$  or  $A = -B$ ; both branches must be followed.

**Question 10** MEN-03 · Pythagoras, mensuration and 3D · spec M5.16

A trapezium has parallel sides of length 8 cm and 14 cm and area  $77 \text{ cm}^2$ . Find the perpendicular distance between the parallel sides.

**Answer: C**

$$\frac{1}{2}(8 + 14)h = 11h = 77 \Rightarrow h = 7.$$

**Full method - how and why**

Trapezium area formula, rearranged for the height.

- Write the formula: area  $= \frac{1}{2}(a + b)h$  where  $a, b$  are the parallel sides.  
why: The height is the **perpendicular** distance between the parallel sides.
- Substitute:  $\frac{1}{2}(8 + 14)h = 11h$ .  
why:  $\frac{1}{2} \times 22 = 11$ .
- Solve:  $11h = 77$ , so  $h = 7$  cm.  
why: Divide both sides by 11.

**Where it goes wrong**

- Forgetting the  $\frac{1}{2}$  and answering 3.5 - an offered option.
- Using the difference of the parallel sides instead of the sum.

**Takeaway** Trapezium: average the parallel sides, then multiply by the perpendicular height.

**Question 11** POL-14 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$  is a cubic polynomial with  $p(1) = p(2) = p(3) = 0$  and  $p(0) = 12$ . Find  $p(4)$ .

**Answer: E**

$$p(x) = a(x-1)(x-2)(x-3); p(0) = -6a = 12 \Rightarrow a = -2. p(4) = -2 \cdot 3 \cdot 2 \cdot 1 = -12.$$

**Full method - how and why**

Three roots determine a cubic up to a constant multiple. One extra value fixes the multiple.

- 1 Write  $p(x) = a(x-1)(x-2)(x-3)$ .

why: Given roots, start from the factorised form; the leading coefficient  $a$  is the only freedom left.

- 2 Use  $p(0) = 12$ :  $a(-1)(-2)(-3) = -6a = 12$ , so  $a = -2$ .

why: Three negative factors give a negative product - a very common sign slip.

- 3 Evaluate  $p(4) = -2(3)(2)(1) = -12$ .

why: Substitute directly into the factorised form; no expansion needed.

**Where it goes wrong**

- Taking  $a = 2$  from  $6a = 12$  and answering  $+12$ .
- Assuming  $a = 1$  because the question says 'a cubic' - the leading coefficient is not given.

**Takeaway** Roots give the shape, one extra point gives the scale; keep everything in factorised form.

**Question 12** ALG-07 · GCSE algebra · spec M4.6

Simplify  $\frac{3}{x+2} - \frac{2}{x-1}$ .

**Answer: D**

$$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = \frac{x-7}{(x+2)(x-1)}.$$

**Full method - how and why**

Subtracting algebraic fractions: common denominator, then expand the numerator carefully - especially the minus sign.

- 1 Common denominator:  $(x+2)(x-1)$ .

why: The denominators share no factor, so their product is the lowest common denominator.

- 2 Rewrite each fraction:  $\frac{3(x-1)}{(x+2)(x-1)} - \frac{2(x+2)}{(x+2)(x-1)}$ .

why: Multiply each numerator by whatever its denominator was missing.

- 3 Combine and expand:  $3x - 3 - (2x + 4) = 3x - 3 - 2x - 4 = x - 7$ .

why: The minus sign applies to **both** terms of the second bracket - the classic error.

- 4 So the answer is  $\frac{x-7}{(x+2)(x-1)}$ .

why: Numerator does not factorise further.

**Where it goes wrong**

- Getting  $x+1$  by only negating the  $2x$  - an offered option.
- Subtracting the denominators as well.

**Takeaway** Bracket the whole second numerator before subtracting, then distribute the minus sign.

**Question 13** CIR-07 · Circles and circle theorems · spec MM3.3

$PQRS$  is a cyclic quadrilateral with angle  $P = 3x$ , angle  $R = x + 40^\circ$ . Find  $x$ .

**Answer: C**

Opposite angles of a cyclic quadrilateral sum to  $180^\circ$ :  $4x + 40 = 180$ ,  $x = 35^\circ$ .

**Full method - how and why**

Cyclic quadrilateral: opposite angles are supplementary. That gives a linear equation immediately.

- 1 Identify the opposite pair: in  $PQRS$  the vertices go round in order, so  $P$  and  $R$  are opposite.

why: Opposite means separated by one vertex on each side - reading the labelling order matters.

- 2 Apply the theorem:  $3x + (x + 40) = 180$ .

why: Opposite angles of a cyclic quadrilateral sum to  $180^\circ$ .

- 3 Solve:  $4x = 140$ , so  $x = 35^\circ$ .

why: Check: angles  $105^\circ$  and  $75^\circ$  sum to  $180^\circ$  ✓.

**Where it goes wrong**

- Setting the angles equal (that is a parallelogram property, not a cyclic one).
- Using  $360^\circ$  and getting  $x = 80^\circ$ .

**Takeaway** Cyclic quadrilateral: opposite angles sum to  $180^\circ$ ; adjacent ones have no such relation.

**Question 14** TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does  $\sin 2x = \frac{1}{2}$  have for  $0 \leq x < 2\pi$ ?

**Answer: C**

$2x$  runs over  $[0, 4\pi)$ , which contains two full periods of  $\sin$ , each giving two solutions: 4 in total.

**Full method - how and why**

Counting solutions of  $\sin(kx) = c$ . Transform the interval for the inside expression and count there.

- 1 Let  $u = 2x$ . As  $x$  runs over  $[0, 2\pi)$ ,  $u$  runs over  $[0, 4\pi)$ .

why: Multiplying the variable by 2 doubles the range of the inside expression - this is the whole trick.

- 2  $[0, 4\pi)$  contains exactly two full periods of  $\sin$ .

why:  $\sin$  has period  $2\pi$ , so a  $4\pi$ -long interval holds two complete waves.

- 3 In each period,  $\sin u = \frac{1}{2}$  has two solutions (one on the way up, one on the way down).

why: A horizontal line strictly between  $-1$  and  $1$  cuts each wave twice.

- 4 Total:  $2 \times 2 = 4$  solutions.

why: Each  $u$  gives exactly one  $x$  via  $x = u/2$ , so no solutions are lost or duplicated.

**Where it goes wrong**

- Answering 2 by counting solutions for  $x$  as if the 2 were not there.
- Answering 8 by doubling twice.

**Takeaway** For  $\sin(kx) = c$  on an interval of length  $L$ , the inside runs over length  $kL$  - count solutions there.

**Question 15** LIN-06 · Straight lines · spec MM3.1

The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

**Answer: A**

Midpoint (4, 3); gradient of segment  $\frac{-4}{6} = -\frac{2}{3}$ , so the bisector has gradient  $\frac{3}{2}$ :  
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$ .

**Full method - how and why**

Perpendicular bisector: through the midpoint, perpendicular to the segment. Two small calculations, then the point-gradient form.

1 Midpoint:  $(\frac{1+7}{2}, \frac{5+1}{2}) = (4, 3)$ .

why: The bisector must pass through the middle of the segment.

2 Gradient of the segment:  $\frac{1-5}{7-1} = -\frac{4}{6} = -\frac{2}{3}$ .

why: Keep the order of subtraction consistent in numerator and denominator.

3 Perpendicular gradient:  $\frac{3}{2}$ .

why: Negative reciprocal of  $-\frac{2}{3}$ .

4 Line through (4, 3):  $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$ .

why: Multiply by 2 to clear the fraction, then collect.

**Where it goes wrong**

- Rearranging to  $3x - 2y = 18$  by a sign slip on the constant.
- Using the segment's gradient instead of the perpendicular one.

**Takeaway** Perpendicular bisector = midpoint + negative reciprocal gradient; it is the locus of points equidistant from the two ends.

**Question 16** LOG-05 · Exponentials and logarithms · spec MM5.2

Solve  $\log_3(x + 6) + \log_3(x - 2) = 2$ .

**Answer: C**

$(x + 6)(x - 2) = 9 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x + 7)(x - 3) = 0$ .  $x = -7$  makes both logs undefined, so  $x = 3$  only.

**Full method - how and why**

Combine the logs, convert to an equation, then **check the domain** - that is what this question is really testing.

1 Combine:  $\log_3[(x + 6)(x - 2)] = 2$ .

why: Sum of logs is the log of the product.

2 Convert to exponential form:  $(x + 6)(x - 2) = 3^2 = 9$ .

why:  $\log_a N = c$  means  $N = a^c$ .

3 Expand and solve:  $x^2 + 4x - 12 = 9 \Rightarrow x^2 + 4x - 21 = 0 \Rightarrow (x + 7)(x - 3) = 0$ , so  $x = -7$  or  $x = 3$ .

why: Ordinary quadratic work.

4 Check both in the **original** equation:  $x = 3$  gives  $\log_3 9 + \log_3 1 \checkmark$ ;  $x = -7$  gives  $\log_3(-1)$ , which is undefined.

So  $x = 3$  only.

why: Logarithms of negative numbers (or zero) do not exist, so roots that break the original expression must be rejected.

**Where it goes wrong**

- Giving both roots - the option ' $x = -7$  or  $x = 3$ ' exists precisely to catch this.
- Checking only one of the two logarithms.

**Takeaway** Always substitute your solutions back: combining logs can create solutions the original equation never had.

**Question 17** LOG-08 · Exponentials and logarithms · spec MM5.3

Given that  $5^x = 2$  and  $2^y = 125$ , find the value of  $xy$ .

**Answer: B**

$$5^{xy} = (5^x)^y = 2^y = 125 = 5^3, \text{ so } xy = 3.$$

**Full method - how and why**

Two exponential facts that chain together. Raising the first to the power  $y$  links them.

- 1 Raise the first equation to the power  $y$ :  $(5^x)^y = 2^y$ , i.e.  $5^{xy} = 2^y$ .

why:  $(a^x)^y = a^{xy}$  - this is what connects the two statements.

- 2 Substitute the second fact:  $2^y = 125$ .

why: Given directly.

- 3 Write 125 as a power of 5:  $5^{xy} = 5^3$ .

why: Both sides are now powers of 5.

- 4 Equate the indices:  $xy = 3$ .

why: Valid because  $5^x$  is one-to-one.

**Where it goes wrong**

- Trying to compute  $x$  and  $y$  separately as logarithms and getting stuck without a calculator.
- Multiplying the two equations to get  $10^{x+y} = 250$ , which leads nowhere.

**Takeaway** To find a **product** of exponents, raise one equation to the other's exponent and chain the results.

**Question 18** LIN-04 · Straight lines · spec MM3.1

The lines  $y = 2x + 1$ ,  $y = -x + 7$  and the  $x$ -axis enclose a triangle. Find its area.

**Answer: B**

$$\text{Intersection: } 2x + 1 = 7 - x \Rightarrow x = 2, y = 5. \text{ } x\text{-intercepts: } -\frac{1}{2} \text{ and } 7, \text{ base} = \frac{15}{2}. \text{ Area} = \frac{1}{2} \cdot \frac{15}{2} \cdot 5 = \frac{75}{4}.$$

**Full method - how and why**

A triangle bounded by two lines and an axis. Find the three vertices, then use the axis segment as the base.

- 1 Find where the two lines meet:  $2x + 1 = -x + 7 \Rightarrow 3x = 6 \Rightarrow x = 2, y = 5$ . Apex (2, 5).

why: Equate the two expressions for  $y$ .

- 2 Find the  $x$ -intercepts:  $2x + 1 = 0$  gives  $x = -\frac{1}{2}$ ;  $-x + 7 = 0$  gives  $x = 7$ .

why: The third side is along the  $x$ -axis, so the other two vertices are the  $x$ -intercepts.

- 3 Base length: from  $-\frac{1}{2}$  to 7 is  $7 - (-\frac{1}{2}) = \frac{15}{2}$ .

why: Subtract, do not add; the negative intercept makes the base longer, not shorter.

- 4 Height is the  $y$ -coordinate of the apex, 5: area  $= \frac{1}{2} \times \frac{15}{2} \times 5 = \frac{75}{4}$ .

why: With a horizontal base, the height is just the apex's distance from the axis.

**Where it goes wrong**

- Using base  $7 - \frac{1}{2} = \frac{13}{2}$  by mishandling the negative intercept.
- Using the apex  $x$ -coordinate 2 as the height.

**Takeaway** Put the base on an axis whenever you can: the height is then simply a coordinate.

**Question 19** CIR-09 · Circles and circle theorems · spec MM3.2

Find the length of the tangent from the point (7, 1) to the circle  $x^2 + y^2 - 2x - 4y - 4 = 0$ .

**Answer: B**

Centre (1, 2), radius 3. Distance from (7, 1) to centre:  $\sqrt{36 + 1} = \sqrt{37}$ . Tangent length =  $\sqrt{37 - 9} = \sqrt{28}$ .

**Full method - how and why**

The tangent, radius and line to the external point form a right-angled triangle. Pythagoras gives the tangent length.

- 1 Put the circle in centre-radius form:  $(x - 1)^2 + (y - 2)^2 = 4 + 1 + 4 = 9$ , so centre (1, 2), radius 3.

why: Complete the square in  $x$  and  $y$ ; the constant  $-4$  moves across.

- 2 Distance from (7, 1) to the centre:  $\sqrt{(7 - 1)^2 + (1 - 2)^2} = \sqrt{36 + 1} = \sqrt{37}$ .

why: This is the hypotenuse of the right-angled triangle (the tangent meets the radius at  $90^\circ$ ).

- 3 Tangent length =  $\sqrt{PC^2 - r^2} = \sqrt{37 - 9} = \sqrt{28}$ .

why: Subtract the squares, not the lengths - the right angle is at the point of contact.

**Where it goes wrong**

- Answering  $\sqrt{37}$  (that is the distance to the centre) or  $\sqrt{37} - 3$ .
- Getting  $r = 9$  instead of 3 from the completed square.

**Takeaway** Tangent length from  $P$ :  $\sqrt{PC^2 - r^2}$ , because tangent  $\perp$  radius makes a right-angled triangle.

**Question 20** BIN-14 · Binomial expansion · spec MM2.4

How many terms in the expansion of  $(x + y + z)^4$  (after collecting like terms) are there?

**Answer: B**

Terms are  $x^a y^b z^c$  with  $a + b + c = 4$ ,  $a, b, c \geq 0$ . Counting: for  $a = 0$  there are 5 choices of  $(b, c)$ ,  $a = 1$ : 4, ...,  $a = 4$ :

1. Total  $5 + 4 + 3 + 2 + 1 = 15$ .

**Full method - how and why**

A trinomial expansion. Counting the distinct terms is a counting problem: how many ways can the powers add to 4?

- 1 Every term has the form  $x^a y^b z^c$  with  $a + b + c = 4$  and  $a, b, c \geq 0$ .

why: Each of the four brackets contributes exactly one of  $x$ ,  $y$  or  $z$ , so the powers add to 4.

- 2 Count systematically by the value of  $a$ : if  $a = 0$  then  $b + c = 4$ , giving 5 pairs;  $a = 1$  gives 4;  $a = 2$  gives 3;  $a = 3$  gives 2;  $a = 4$  gives 1.

why: For a fixed  $a$ , the number of  $(b, c)$  pairs is  $(4 - a) + 1$ .

- 3 Total:  $5 + 4 + 3 + 2 + 1 = 15$ .

why: A triangular number - the standard result  $\binom{n+2}{2}$  for three variables, here  $\binom{6}{2} = 15$ .

**Where it goes wrong**

- Answering  $3^4 = 81$  - that counts the terms **before** collecting like terms.
- Answering 5 by counting only the powers of  $x$ .

**Takeaway** Distinct terms of  $(x + y + z)^n$  correspond to solutions of  $a + b + c = n$  in non-negative integers:  $\binom{n+2}{2}$  of them.